

Query/Retrieve Use Cases

Version 1.1

Use Cases

New Patient

1. Get all images related to Patient

Existing patient

1. Get any new images related to Patient.
 - a. ALL
 - b. Specific
 - i. Only want the PET that was done last week.
2. I want the current plan for patient A which includes structure set and dose
3. I want to find a plan/images/rt structures/dose that was delivered last week Thursday.
4. I want to see all of Dr. Jones's patient.
5. I want to see all plans for this body site. If I'm treating the pelvis I want would want to know if there were any prior treatments that are pelvis or near the pelvis.
6. *for discussion* My machine went down and I want to know which patient is assigned to this machine.
7. I want to know which plan was approved.
8. I want to know all plans for John that has a dose of 4500 cGy prescribed to the target. What was the final dose and total dose to the organ at risk.
9. I want know all plans with fractionation with fewer than 5 fractions so that I know which plans I am delivering large amount of dose. I like to see the target prescription dose as well. So I know how many fractions for which dose.
10. Does he have any restriction on body.
11. Why is he coming back.

From IHE-wiki:

- Retrieve images, structures, plan from CTSim/Archive for planning
- Retrieve secondary images from an archive or a modality to assist in planning
- Retrieve plan and doses from other planning systems for comparison
- Retrieve treatment records and verification images for offline review
- Retrieve plan and dose for dose accumulation
- Retrieve a complete set of patient treatment planning data for clinical trials submission

Assumptions

The baseline behavior of the DICOM Query/Retrieve C-FIND operation requires a hierarchical search method. This means that in order to find a certain Series the SCU must do two prior queries to find the previous levels namely Study and Patient. And three for Composite Object Instance queries.

The standard defines an extended relational query behavior for C-FIND SCU and SCP that enables applications to skip the Study and Patient level queries and directly search for Series. This however requires the query to contain the unique keys (Patient ID and Study Instance UID). Some SCP implementations might support universal matching, which means that the SCU must include those two attributes in the query dataset but might leave the Value empty or set to '*'. This also applies to the Composite Object Instance level.

The assumption is that the intended use-case can be achieved using either baseline behavior and relational queries. The latter method will minimize the number queries needed, but bottom line the result will be the same.

Proposed flow

New Patient

1. Get all images related to Patient

Existing patient

1. Get any new images related to Patient.
 - a. ALL
 - b. Specific
 - i. Only want the PET that was done last week.
2. I want the current plan for patient A which includes structure set and dose

Main issue here is that the dose references the plan and not the other way around. The Content Receiver must therefore query for all plans before querying for doses that references those plans.

Option 1 – Find all plans using various C-FIND on RTSTRUCT/RTPLAN/RTDOSE:

Subtask 1: Find all patients structure set instances

1. C-FIND SERIES (all structure set series)
Modality = 'RTSTRUCT'
2. C-FIND INSTANCE (structure set instances)
SOP Class UID = RT Structure Set Storage
Series Instance UID = previous result

Subtask 2: Find all patients plans

Commented [SPB1]: Content Retriever sends C-FIND with match key Modality (0008,0060) = 'CT\MR\PT'

This is tricky. Even if the modality depicts an image modality, the actual data might contain something else. Such as raw data. The only way to know for sure that a given Series is a real image is using the SOP Class UID, which is a Composite Instance level attribute. More defined terms can be added for additional modalities,

No additional requirements from profile needed. Default SCU/SCP behavior enough.

Commented [SPB2]: No additional requirements from profile needed. Default SCU/SCP behavior enough.

Commented [SPB3]: What does current really mean? Is it already available to the operator or is it the latest plan for patient A?
Does this mean 'the plan has structure set and dose' or 'I want structure set, plan and dose'?

3. C-FIND SERIES (referenced plan series)
Modality (0008,0060) = 'RTPLAN'
Referenced RT Plan Sequence (300C,0002) = result from subtask 1
4. C-FIND INSTANCE (plan instances)
Series Instance UID = previous result
SOP Class UID = RT Plan Storage

Subtask 3: Find patients doses

5. C-FIND SERIES (all dose series)
Modality = 'RTDOSE'
6. C-FIND INSTANCE (dose instances)
Series Instance UID = previous result
SOP Class UID = RT Dose Storage
Referenced RT Plan Sequence (300C,0002) = patient plans (subtask 2)
Dose Summation Type = x
Dose Type = x

Subtask 4: Process result

7. Order plans by Content Date/Time and return first

Option 2 – Use RT Plan Geometry:

Subtask 1: Find all patients non-geometric plans

1. C-FIND SERIES (plan series)
Modality (0008,0060) = 'RTPLAN'
RT Plan Geometry (300A,000C) = 'PATIENT'
2. C-FIND INSTANCE (plan instances)
Series Instance UID = previous result

Subtask 2: Find patients doses

3. C-FIND SERIES (all dose series)
Modality = 'RTDOSE'
4. C-FIND INSTANCE (dose instances)
Series Instance UID = previous result
Referenced RT Plan Sequence (300C,0002) = patients plans (subtask 2)

Subtask 3: Process result

5. Filter and order plans by Content Date/Time and return first

Option 3 – Use C-FIND-REFERENCED

Subtask 1: Find all patients plans

1. C-FIND SERIES
Modality = 'RTPLAN'
2. C-FIND INSTANCE
SOP Class UID = RT Plan Storage
Series Instance UID = previous result
3. C-FIND-RELATED
SOP Instance UID = previous result
4. Filter and order plans by Content Date/Time and return first

3. I want to find a plan/images/rt structures/dose that was delivered last week Thursday.
4. I want to see all of Dr. Jones's patient.
5. I want to see all plans for this body site. If I'm treating the pelvis I want would want to know if there were any prior treatments that are pelvis or near the pelvis.

Main issue here is fuzzy matching. Treatment site is not well defined in RT Plan.

Option 1 – Match by Treatment Sites (300A,000B)

1. C-FIND SERIES (all plans)
Modality = 'RTPLAN'
2. C-FIND INSTANCE
Series Instance UID = previous result
Treatment Sites (300A,000B) = '*pelvis*'

Option 2 – Match by Prescription Target (ROI)

Option 3 – Match by Beam Isocenter

Commented [SPB4]: Query for treatment records?

Commented [SPB5]: Is this Referring Physician's Name in General Study?

Commented [SPB6]: This requires fuzzy matching and needs to be negotiated with SCP. Some SCP implementers might use fussy matching by default. Otherwise, case-sensitivity and localization will be a problem.

Commented [SPB7]: Just added for completion. Option 2 would match the prescription (Dose Reference) Option 3 (advanced) would match the beam isocenters against the volume of target structures.

6. *for discussion* My machine went down and I want to know which patient is assigned to this machine.
7. I want to know which plan was approved.

C-FIND INSTANCE Approval Status = 'APPROVED'

8. I want to know all plans for John that has a dose of 4500 cGy prescribed to the target. What was the final dose and total dose to the organ at risk.

C-FIND INSTANCE Target Prescription Dose = 4500

9. I want to know all plans with fractionation with fewer than 5 fractions so that I know which plans I am delivering large amount of dose. I like to see the target prescription dose as well. So I know how many fractions for which dose.

C-FIND INSTANCE Number Of Fractions Planned = '0\1\2\3\4\5'

Commented [SPB8]: Range matching only supported for dates. This could be extended to support number range matching (e.g. 0-5).

10. Does he have any restriction on body.

Commented [SPB9]: Is this HIS information? Like patient flags?

11. Why is he coming back.

Commented [SPB10]: Not sure how to translate into C-FIND. Previous diagnosis?